

DANISH SILVA

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CAREER OBJECTIVE

Final-year Mechatronics Engineering student at UTS with team leadership experience, working toward a career as a robotics engineer. I'm keen for graduate roles spanning embedded systems, electronics, computer vision, and mechanical design. My core strength is hands-on electronics and embedded work, taking hardware from schematic and PCB design through power, sensing, and control to a tested, working system, backed by strong mechanical CAD skills. I'm motivated to bring these disciplines together and grow into an industry-ready robotics engineer.

EDUCATION

Bachelor of Engineering (Honours)

Feb 2024 - Jun 2027 (Expected)

University Of Technology Sydney

- Mechatronics Engineering - 80 WAM
- Sub-Major: Applied and Industrial Optimisation BCS Hons

WORK EXPERIENCE

Engineering Intern & Team Lead

May 2026 - Aug 2026

Optik Consultancy — Client: Sensear, Sydney

- **Team Leadership:** Led a 7-member engineering team building an in-house EN 352 compliance testing system for Sensear's industrial earmuff headsets, replacing costly external lab testing, while owning project direction, task allocation, and client engineering communication.
- **Power & Control Electronics:** Designed the test-system electronics end-to-end, including ESP32-based control, a high-torque servo drive for headband-fatigue cycling, and a multi-rail power supply with buck and linear regulation, plus reverse-polarity, over-current, and over-voltage protection.
- **Sensor Front-End:** Engineered the headband-force measurement front-end, using a Wheatstone-bridge load cell with a 24-bit ADC, differential filtering, and ratiometric excitation, for repeatable clamp-force measurement to EN 13819-1.
- **System Design:** Delivered full system schematics, a custom ESP32-S3 PCB, and a costed bill of materials, and specified the acoustic measurement chain (loudspeaker array and audio interface) for attenuation testing per ISO 4869.

Marketing Coordinator

Mar 2022 - Jun 2024

LRS Australia, North Sydney

- Coordinated design, logistics, and on-site operations across 3+ national expos with internal teams and external vendors, increasing customer leads.
- Diagnosed and resolved hardware and software faults through systematic testing (90%+ customer satisfaction), and designed a 3-step dispatch verification checklist that reduced shipping errors.
- Liaised with customers, printers, and vendors to deliver high-quality outcomes under tight deadlines, strengthening communication, stakeholder management, and adaptability.

PROJECTS

MEMS-Based Step Tracker

Feb 2026 - Jun 2026

- Designed a full custom PCB shield for an ESP32-based step tracking device in Altium Designer, responsible for end-to-end power supply unit (PSU) design and system-level PCB integration.
- Engineered a 3.3V LDO-based power regulation circuit incorporating over-current protection (OCP) and over-voltage protection (OVP), pre-validated through LTSpice simulation to verify current draw and protection threshold behaviour prior to fabrication.
- Integrated and assembled the complete PCB, bringing together independently developed subsystems including a MEMS accelerometer, analogue signal conditioning filter, and peripheral interfaces into a unified, functional board.
- Managed inter-subsystem compatibility at the schematic and layout level, ensuring correct signal integrity, power domain separation, and component placement across the full board.

UR3e RealSense Visual Servoing

Aug 2025 - Nov 2025

- Developed an eye-in-hand Image-Based Visual Servoing (IBVS) system for a UR3e robotic arm, integrating an Intel RealSense D435 RGB-D camera with ROS for closed-loop visual control.
- Implemented real-time checkerboard detection and depth estimation using OpenCV, calculating image-space error and generating camera velocity commands through the IBVS control law and interaction matrix.
- Converted camera-space velocity into joint velocities using the tool-to-camera transformation and UR3e Jacobian pseudoinverse, enabling the robot to autonomously track and maintain alignment with a moving target.
- Developed a real-time ROS control pipeline using rosbridge and TCP communication, streaming micro-step joint trajectories at 15–20 Hz while tuning controller gain and velocity limits to achieve stable, responsive motion.

SKILLS

- **Programming:** Python, C, C++
- **Embedded & Systems:** Arduino, Raspberry Pi, ESP32, STM32, Intel Quartus (FPGA)
- **PCB & Electrical Design:** Altium Designer, KiCad
- **Robotics Frameworks:** ROS2, OpenCV, MoveIt 2
- **Mechanical Design:** SolidWorks, Autodesk Fusion, AS1100 Technical Drawings
- **Simulation:** Gazebo
- **Data & Optimisation:** MATLAB, NumPy
- **Version Control:** GitHub, Altium365
- **Core Skills:** Leadership, Teamwork, Adaptability, Analytical Thinking, Process Optimisation, Stakeholder Management, Technical Documentation

REFERENCES

Available upon request